

dice simulator 155
 dictionaries 135
 direction 39
 directories **193**, 218
 directory trees 193
 division 52, 102, 112
 domain names 194
 drawing
 on a canvas 157, 165
 with coordinates 158
 pens and turtles 48–9, 122, 152
 Drawing machine project 140–7
 drums 59

E
 educational apps 206
 electrical signals 195
 “elif” command 121
 “else-if” command *see* “elif”
 command
 emails
 spam 213
 websites 198, 208
 encryption **202**, 218
 “end” label 117
 errors 94–5
 finding and fixing 148–9
 Escape the dragon! project 32–7
 event handlers 166, 211
 events 211, 218
 reacting to **162–3**
 “Events” blocks 31, 32, **44–5**
 EXCLUSIVE OR gate 187
 experimenting
 with coding 19
 with Python 176–7
 with Scratch 82–3

F
 Facebook 201
 false *see* true or false
 fiberoptic cables 195
 file browsers 190
 files 218
 compressing 202
 corruption 213
 managing 193
 properties 192
 sizes 192
 storing data in **192–3**
 firewalls 213

floats **110**, 218
 flowcharts **141**, 146
 folders 193
 “for” loops 122, 124
 “forever” block 23, 31, 32, 33, 38,
 39, 46, 47, 125
 forever loop 46, 47, 69, 103, 125,
 133
 Fortran 199
 full stops 117
 functions 72, **130–1**, 132–3, 218
 calling 104, **130**, **139**
 getting data back from 131
 making 130
 passing data to 131
 using and naming 143
 and variables 138–9

G
 gadgets 215
 Game Boy 201
 game engines 205
 game logic 205
 games 14, **204–5**
 customizing 81
 improving 174
 in JavaScript 209
 making with Python 177
 for mobile devices 206
 online 195
 random numbers in 53
 see also projects
 Gates, Bill 200
 gates, logic **186–7**, 218
 Ghost game project 96–9
 gigabytes (GB) 189, 192
 global variables 138, 139
 GNU/Linux operating system 201
 Google 86, 201, 217
 GPU 218
 graphic designers 204
 graphical user interface
 see GUI
 graphics 218
 changing 160–1
 color and coordinates 156–7
 effects 43, 152, 153
 games 205
 making shapes 158–9
 graphics processing unit
 see GPU
 grid, x and y 56, 158
 GUI 154–9, 218

H
 hackers **213**, 218
 hardware 15, 181, 191, 218
 programming 198
 hexadecimal 156, 183, 185, 218
 high-level languages 191
 Hopper, Grace 200
 HTML (HyperText Markup
 Language) 208, 209, 210

I
 IDLE
 code and shell windows 106–7
 color codes 107
 and errors 148
 how it works 93
 and installation of Python
 89, 90, 91
 what is IDLE? **88**
 working in **92–3**, 209
 “if” block/command 64, 120
 “if-elif-else” command 121
 “if-else” command 120
 “if-then” block/command 64, 66,
 67, 103
 “if-then-else” block/command
 64, 65, 76, 103
 “in” operator 119
 indents 98
 index numbers 115, 128, 218
 infinite loops 125
 input
 blocks with 73
 devices 162, 180, 189
 logic gates 186–7
 program 31, 100, 101, 106,
 116, **180**, 218
 “input()” function 101, 130
 insertion sort 150
 installation
 Python 3 88–91
 Scratch 24–5
 instruments, musical 59
 integers **110**, 219
 Integrated DeveLopment
 Environment *see* IDLE
 interface
 GUI 154–5
 Scratch **26–7**, 49
 user 140, 146–7, 219
 Internet **194–5**
 connecting to other
 computers via 152

 connection to 19
 programming for **208–9**
 Internet browsers 209
 Internet Protocol 194
 interpreters 191
 inverters 186
 iOS devices 198
 IP addresses **194**, 195, 219

J
 Java 18, 198
 JavaScript 18, 198, 208, 209
 using **210–11**
 JPEGs 193

K
 keyboard
 events 44, 66, 162, 163, 211
 input function 116, 162, 163,
 180
 keys, dictionaries 135
 kilobytes (KB) 183, 192

L
 laptops 181
 latch circuits 189
 letter commands 141
 level designers 204
 libraries 219
 building your own 176
 help and documentation
 153
 importing 153
 light, data transmission as
 pulses of 195
 Linux 88, 201
 lists
 adding/deleting items 55, 105,
 128, 129, 169
 combining 129
 commands 105
 copying 137
 creating 55, 105
 in loops 129
 playing with 55
 putting tuples in 134
 in Python **128–9**, 132–3
 in Scratch **54–5**
 using 55, 129, 167
 in variables **136–7**